

Solution Report

Implementation Details

We implemented a generalized version of the **Yannakakis algorithm** to evaluate k -line join queries of the form:

$$q(A_1, \dots, A_{k+1}) : -R_1(A_1, A_2), R_2(A_2, A_3), \dots, R_k(A_k, A_{k+1})$$

The implementation in [Solution.js](#) handles arbitrary values of k (tested up to $k = 9$) using a dynamic approach rather than hardcoded steps.

Algorithm Steps

1. **Semijoin Reduction Phase:** To ensure **global consistency**, we perform two passes of semijoin reductions.

- **Right-to-Left Pass:** For $i = k - 1$ down to 1, we reduce R_i by R_{i+1} :

$$R'_i = R_i \bowtie R_{i+1}$$

This propagates constraints from the end of the chain to the beginning.

- **Left-to-Right Pass:** For $i = 2$ up to k , we reduce R_i by R_{i-1} :

$$R''_i = R'_i \bowtie R'_{i-1}$$

This propagates constraints from the beginning back to the end.

After these two passes, all relations are fully reduced, meaning every tuple in every relation participates in at least one valid result tuple.

2. **Join Phase (DFS):** We perform a Depth-First Search (DFS) to enumerate the results.

- We start by iterating through the tuples of the first relation R_1 .
- For a tuple in R_i , we use a hash index to efficiently find matching tuples in R_{i+1} .
- We recursively build the result tuple $(A_1, \dots, A_{k+1})$.
- Because of the reduction phase, we are guaranteed that every path in the DFS leads to a valid result (no dead ends).

Complexity Analysis

The exact complexity of the algorithm is $O(N + OUT)$. This ensures the solution meets the requirement of linear behavior with respect to the input size N and output size OUT .

1. **Reduction Phase:**

- We perform $2(k - 1)$ semijoin operations.
- Each semijoin $R \bowtie S$ takes $O(|R| + |S|)$ time using hash indices.

- Total time for reduction is proportional to the total size of all input relations: $O(N)$.

2. Join Phase:

- The DFS traversal visits each result tuple exactly once.
- Since the relations are globally consistent, there is no wasted work exploring invalid paths.
- The time spent is proportional to the number of output tuples: $O(OUT)$.

Total Complexity: $O(N + OUT)$.

Execution Results

We ran the generalized algorithm with three test cases:

Case 1: k=3 (Original Example)

- **Input:** 3 relations with 5 tuples each (plus noise).
- **Result:** Found **4 tuples**.
- **Output Sample:**

(index)	A1	A2	A3	A4
0	1	10	100	1000
1	2	20	200	2000
2	2	20	200	2001
3	4	40	400	4000

Case 2: k=5

- **Input:** 5 relations with 10 tuples each (generated chain with noise).
- **Result:** Found **15 tuples**.
- **Output:**

(index)	A1	A2	A3	A4	A5	A6
0	0	0	0	0	0	0
1	1000	1000	1000	1000	1000	1000
2	1	1	1	1	1	1
3	2	2	2	2	2	2
4	1002	1002	1002	1002	1002	1002
5	3	3	3	3	3	3
6	4	4	4	4	4	4
7	1004	1004	1004	1004	1004	1004
8	5	5	5	5	5	5
9	6	6	6	6	6	6
10	1006	1006	1006	1006	1006	1006
11	7	7	7	7	7	7
12	8	8	8	8	8	8
13	1008	1008	1008	1008	1008	1008
14	9	9	9	9	9	9

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Case 3: k=9

- **Input:** 9 relations with 20 tuples each (generated chain with noise).
- **Result:** Found **30 tuples**.
- **Output:**

(index)	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10
0	0	0	0	0	0	0	0	0	0	0
1	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000
2	1	1	1	1	1	1	1	1	1	1
3	2	2	2	2	2	2	2	2	2	2
4	1002	1002	1002	1002	1002	1002	1002	1002	1002	1002
5	3	3	3	3	3	3	3	3	3	3
6	4	4	4	4	4	4	4	4	4	4
7	1004	1004	1004	1004	1004	1004	1004	1004	1004	1004
8	5	5	5	5	5	5	5	5	5	5
9	6	6	6	6	6	6	6	6	6	6
10	1006	1006	1006	1006	1006	1006	1006	1006	1006	1006
11	7	7	7	7	7	7	7	7	7	7
12	8	8	8	8	8	8	8	8	8	8
13	1008	1008	1008	1008	1008	1008	1008	1008	1008	1008
14	9	9	9	9	9	9	9	9	9	9
15	10	10	10	10	10	10	10	10	10	10
16	1010	1010	1010	1010	1010	1010	1010	1010	1010	1010
17	11	11	11	11	11	11	11	11	11	11
18	12	12	12	12	12	12	12	12	12	12
19	1012	1012	1012	1012	1012	1012	1012	1012	1012	1012
20	13	13	13	13	13	13	13	13	13	13
21	14	14	14	14	14	14	14	14	14	14
22	1014	1014	1014	1014	1014	1014	1014	1014	1014	1014
23	15	15	15	15	15	15	15	15	15	15
24	16	16	16	16	16	16	16	16	16	16
25	1016	1016	1016	1016	1016	1016	1016	1016	1016	1016
26	17	17	17	17	17	17	17	17	17	17
27	18	18	18	18	18	18	18	18	18	18
28	1018	1018	1018	1018	1018	1018	1018	1018	1018	1018
29	19	19	19	19	19	19	19	19	19	19

The algorithm successfully scaled to handle longer chains, correctly filtering out dangling tuples and producing the expected join results.